03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 22

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SUMMARY

All equations of stellar structure

$$\text{Mass equation: } \frac{dm}{dr} = 4\pi r^2 \rho$$

$$\text{Hydrostatic equilibrium: } \frac{dP}{dr} = -\frac{G\, m\rho}{r^2}$$

$$\text{Energy generation: } \frac{dL}{dr} = 4\pi\, r^2 \left[\rho\epsilon - \rho \frac{d}{dt}\left(\frac{u}{\rho}\right) + \frac{P}{\rho}\frac{d\rho}{dt}\right]$$

$$\text{Radiation transport: } \frac{dT}{dr} = -\frac{3\,\kappa\rho\, L}{16\pi\, a\, c\, r^2\, T^3}$$

$$\text{Convection transport: } \left(\frac{dT}{dr}\right)_{\text{ad}} = \frac{\Gamma_2 - 1}{\Gamma_2}\frac{T}{P}\frac{dP}{dr}$$

$$\text{Chemical evolution: } \frac{dX}{dt} = r_X \text{ etc.}$$

Additional essential relations

- The dependent variables in these equations are m or r, P, L, T, X [and other chemical mass fractions]
- We thus need expressions for $\rho, \epsilon, u, \kappa, \Gamma_2, r_X$ in terms of pressure, temperature and chemical compostion.
- We've seen these throughout the lectures, eg
  - Pressure, density, temperature relations - lecture 4
  - Mean molecular weight - lecture 4
  - Density models, eg linear, quadratic, polytropes - lecture 6
  - Opacity - lecture 8
  - Energy generation rate - lecture 11/12

The stellar evolution timescales - lecture 1

- Three main timescales:
  - Dynamical: $t_{\mathrm{dyn}} = \sqrt{\frac{2R^3}{GM}}$
  - Thermal: $\tau_{th} = \frac{GM_*^2}{R_* L}$
  - Nuclear: $t_{\mathrm{nuc}} = 7 \cdot 10^{-4} \frac{M_* c^2}{L_*}$
- $t_{\mathrm{dyn}} < \tau_{th} < t_{\mathrm{nuc}}$
- Use scaling relations with the Sun to make your life easier

Stellar modeling assumptions - lecture 2

- Main assumptions:
  - Spherical symmetry
  - Hydrostatic equilibrium
  - Thermodynamic equilibrium
  - Only convection transports chemical elements
  - Ignore: rotation, magnetic field, layer mixing, mass-loss
- Gas is usually assumed to be fully described by its dominating properties, ie radiation pressure, ideal gas, or electron degeneracy

Radiation versus convection - lectures 7-10

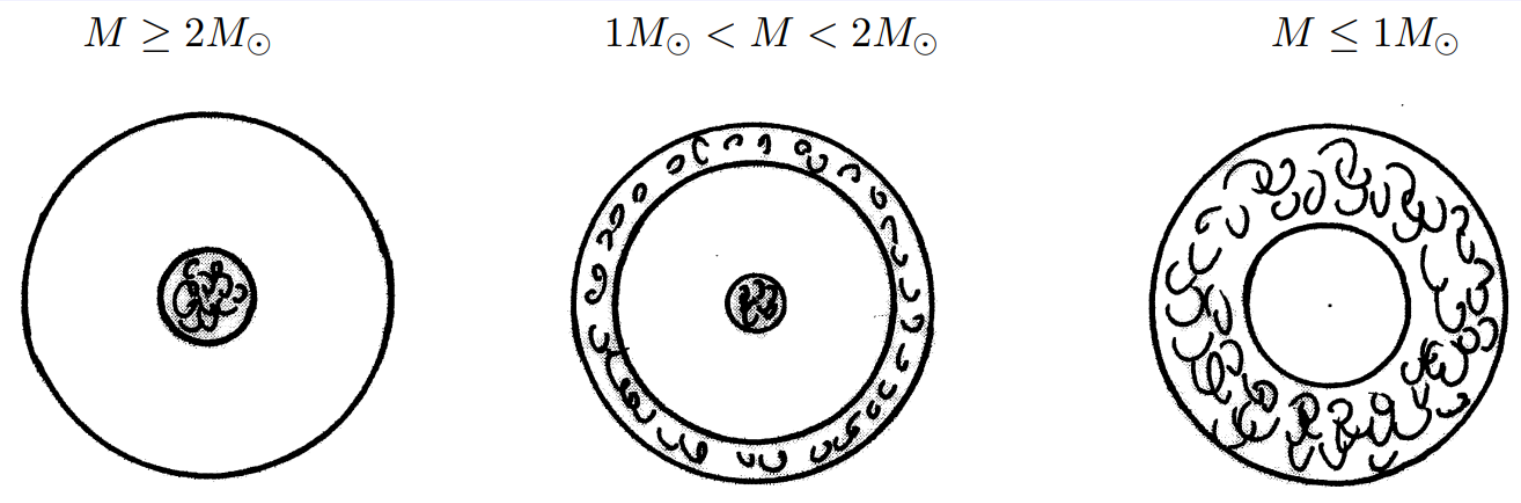

Image credit: JCD - Fig 6.2

- There are radiative and convective zones. The structure is mainly dependent on stellar mass.
- Note: Some very cool stars (cool M dwarfs) can be fully convective.
- In each zone, energy is transported differently, often described with a temperature gradient.
- We assume throughout that convection is the only method to mix chemical elements as well.

Mass-luminosity relation - lecture 10

- Lower Main sequence: $L_{*,\text{low mass MS}} \propto M_*^5 \mu^{7.5}$
- Upper main sequence: $L_{*,\text{high mass MS}} \propto M_*^3 \mu^4$
- Fully convective stars: $L_{*,\text{convective}} \propto M_*^{-7} T_{\text{eff}}^{40}$

Nuclear energy generation - lecture 11-13

- Composition described with X, Y, Z where $X + Y + Z = 1$
- Change in composition described with reaction rate e.g.

$$\frac{dX}{dt} = r_X$$

- This depends on the cross section, which requires knowledge of the Gamow peak
- Nuclear fusion per element:
  - Hydrogen burning:
    - Proton-proton chain - 3 ways
    - CNO cycle
  - He burning through triple alpha process
  - Next elements to fuse: C, Ne, O, "Si"

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The HR diagram

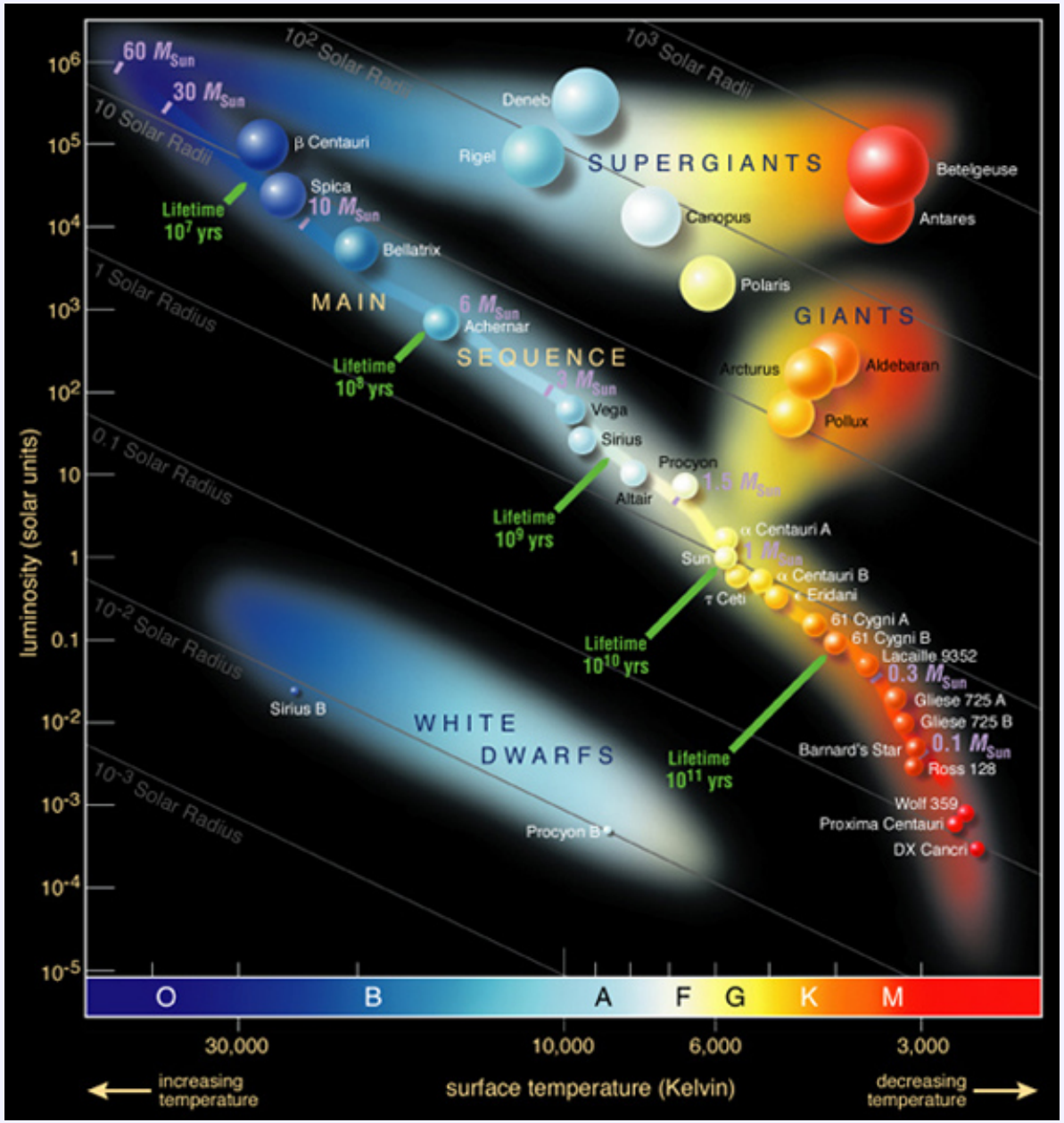

Image credit: ESO

The Gaia colour-magnitude diagram (observed HR)

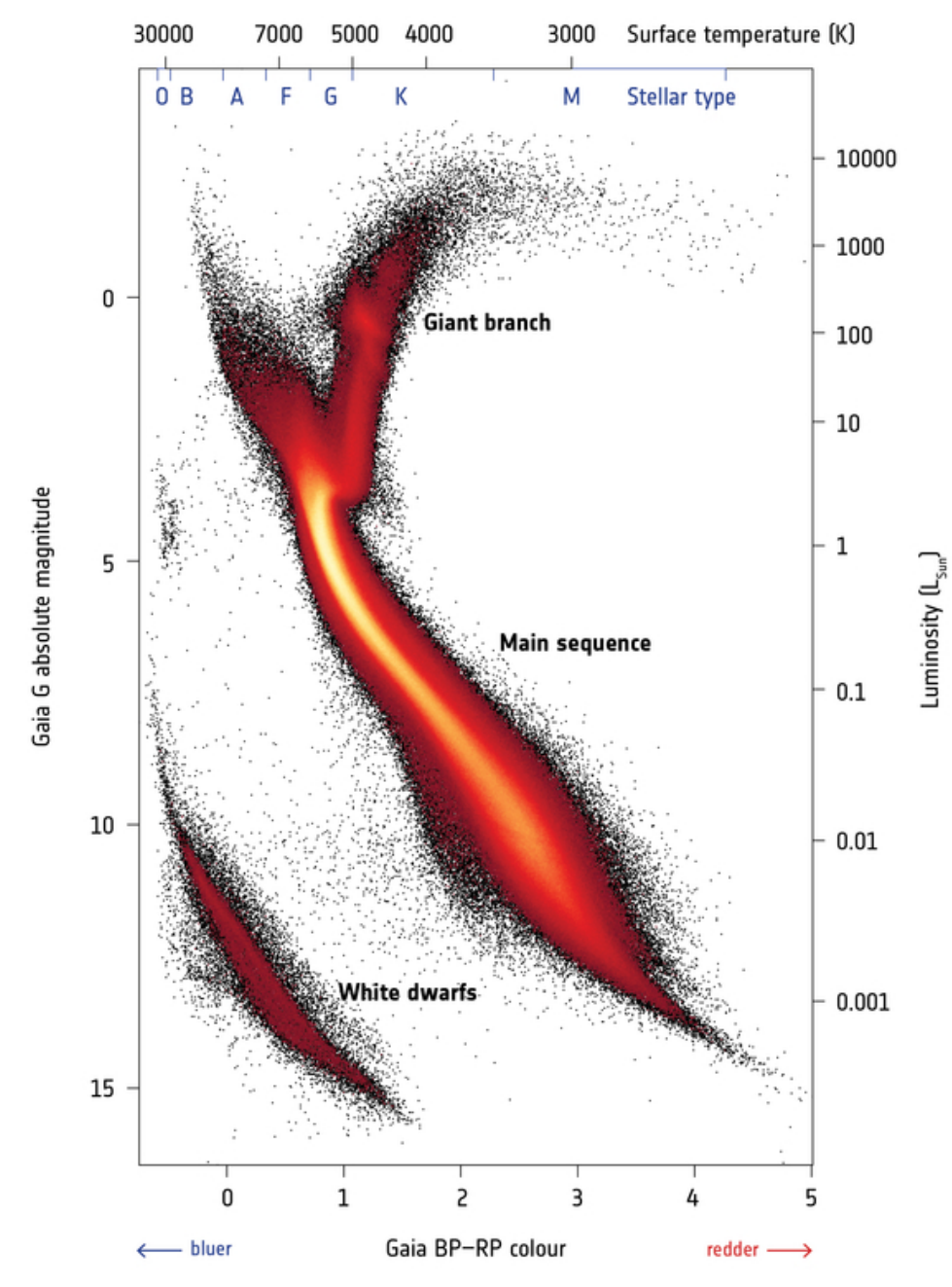

Image credit: ESA

Stellar evolution

- Pre-main sequence = stage before core H fusion starts (lecture 14)
- Main-sequence = stage during core H fusion (lecture 15)
- Hayashi tracks = tracks for fixed effective temperature where stars usually have deep convective envelopes (lecture 10)
- Forbidden region = region with effective temperatures colder than the Hayashi Tracks (lecture 10)
- Stellar evolution is mass-dependent:
  - Low-mass (below 1-2 solar masses) - lecture 18
  - Moderate-mass (between 1-2 and 8-10 solar masses) - lecture 16-17
  - High-mass (between 8-10 and 20-30 solar masses) - lecture 20
  - Highest mass (above 20-30 solar masses) - lecture 20

Stellar evolution features / concepts

- "Hook" after the MS
- Hertzsprung gap
- He flash
- Red Giant branch
- Asymptotic giant branch
- Horizontal branch
- Schönberg-Chandrasekhar limit
- Chandrasekhar limit

Four end states of stars - lecture 21

- White Dwarf: end state for low-mass and moderate-mass stars (lower than 8-10 Solar masses)
- Neutron Star: end state after a core-collapse supernova (type Ib/c or II), for massive stars
- Black Hole: end state of
  - supermassive stars (higher than 20 Solar masses)
  - neutron stars who accreted too much material
- Nothing: end state of white dwarf that accreted too much material after which it explodes through a Supernova type Ia

What is next?

- A well-deserved break for all of us
- Revision lecture on Thursday 7th May 2026 at 10am. It will take place in Arts Lecture Room 7 - 223.
- Exam - 1.5hr, formula sheet on Canvas, no question choice
- Another well-deserved break :)

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